Case Study 4: Multi-Day Water Tank Simulation

Aim: To develop a Python program that simulates water tank levels over multiple days by processing changing inflow and outflow quantities, updating water levels, detecting overflow and shortages, and identifying the first day on which the water level reaches a specified threshold.

1. Problem Statement

A water tank receives water from different sources and supplies water for daily usage. The water level changes depending on the amount of inflow and outflow.

The program should:

1. Accept the initial water level and tank capacity.

2. Accept the number of days.

3. Read daily water inflow and outflow quantities.

4. Update the water level every day.

5. Detect overflow conditions.

6. Detect water shortages.

7. Identify the first day when the water level reaches a specified threshold.

8. Display the final water level and daily report.

9. Assumed Conditions

Parameter	Value
Tank capacity	1000 litres
Initial water level	500 litres
Overflow	Water exceeds 1000 litres
Shortage	Water demand exceeds available water
Threshold	800 litres

3. Algorithm

4. Start.

5. Enter tank capacity.

6. Enter initial water level.

7. Enter the number of days.

8. Enter the target water threshold.

9. For each day:

   - Read inflow quantity.

   - Read outflow quantity.

   - Calculate available water.

   - Detect overflow.

   - Detect shortage.

   - Update the water level.

   - Check whether the threshold is reached for the first time.

7. Display the daily water level.

8. Display overflow and shortage details.

9. Display the first threshold-reaching day.

10. Display the final water level.

11. Stop.

12. Pseudocode

START

INPUT tank_capacity

INPUT initial_water_level

INPUT number_of_days

INPUT threshold

SET water_level = initial_water_level

SET first_threshold_day = 0

```
FOR day = 1 TO number_of_days

    INPUT inflow
    INPUT outflow

    available_water = water_level + inflow

    IF available_water > tank_capacity
        overflow = available_water - tank_capacity
        available_water = tank_capacity
    ELSE
        overflow = 0
    END IF

    IF outflow > available_water
        shortage = outflow - available_water
        water_level = 0
    ELSE
        shortage = 0
        water_level = available_water - outflow
    END IF

    IF water_level >= threshold AND first_threshold_day = 0
        first_threshold_day = day
    END IF
```

DISPLAY water_level, overflow, shortage

END FOR

DISPLAY final water level

DISPLAY first threshold day

STOP

5. Python Program

Complete Python solution

Multi-Day Water Tank Simulationcapacity = float(input("Enter tank capacity (litres): "))water_level = float(input("Enter initial water level (litres): "))days = int(input("Enter number of days: "))threshold = float(input("Enter target water threshold: "))# Initialize counterstotal_overflow = 0total_shortage = 0overflow_days = 0shortage_days = 0first_threshold_day = -1daily_records = []# Process each dayfor day in range(1, days + 1): print("\nDay", day) inflow = float(input("Enter water inflow: ")) outflow = float(input("Enter water outflow: ")) if inflow < 0 or outflow < 0: print("Invalid quantity! Enter non-negative values.") continue # Calculate available water available_water = water_level + inflow

6. Sample Output

Enter tank capacity (litres): 1000

Enter initial water level (litres): 500

Enter number of days: 4

Enter target water threshold: 800

Day 1

Enter water inflow: 400

Enter water outflow: 200

Water level: 700 litres

Overflow: 0 litres

Shortage: 0 litres

Day 2

Enter water inflow: 500

Enter water outflow: 100

Water level: 900 litres

Overflow: 100 litres

Shortage: 0 litres

Day 3

Enter water inflow: 100

Enter water outflow: 1100

Water level: 0 litres

Overflow: 0 litres

Shortage: 100 litres

Day 4

Enter water inflow: 500

Enter water outflow: 100

Water level: 400 litres

Overflow: 0 litres

Shortage: 0 litres

========== WATER TANK REPORT ==========

Day Inflow Outflow Level Overflow Shortage

1 400 200 700 0 0

2 500 100 900 100 0

3 100 1100 0 0 100

4 500 100 400 0 0

Final Water Level: 400 litres

Total Overflow: 100 litres

Total Shortage: 100 litres

Overflow Days: 1

Shortage Days: 1

First Threshold Reached on Day: 2

========== SIMULATION COMPLETED ==========

7. Flowchart

#chatgpt-mermaid-_r_261_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_261_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_261_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_261_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-

mermaid-_r_261_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_261_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_261_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_261_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_261_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_261_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_261_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_261_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_261_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_261_ p{margin:0;}#chatgpt-mermaid-_r_261_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_261_ .label text,#chatgpt-mermaid-_r_261_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ .node rect,#chatgpt-mermaid-_r_261_ .node circle,#chatgpt-mermaid-_r_261_ .node ellipse,#chatgpt-mermaid-_r_261_ .node polygon,#chatgpt-mermaid-_r_261_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_261_ .rough-node .label text,#chatgpt-mermaid-_r_261_ .node .label text,#chatgpt-mermaid-_r_261_ .image-shape .label,#chatgpt-mermaid-_r_261_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_261_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_261_ .rough-node .label,#chatgpt-mermaid-_r_261_ .node .label,#chatgpt-mermaid-_r_261_ .image-shape .label,#chatgpt-mermaid-_r_261_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_261_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_261_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_261_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_261_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_261_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_261_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_261_ .edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_261_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_261_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_261_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_261_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_

div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_261_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_261_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_261_ .icon-shape,#chatgpt-mermaid-_r_261_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_261_ .icon-shape p,#chatgpt-mermaid-_r_261_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_261_ .icon-shape .label rect,#chatgpt-mermaid-_r_261_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_261_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_261_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_261_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_261_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_261_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_261_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_261_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_261_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_261_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_261_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_261_ [data-look="neo"].node .outer-path{filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_261_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_261_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_261_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_261_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_261_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_261_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_261_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_261_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_261_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_261_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_261_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_261_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_261_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_261_ .node rect,#chatgpt-mermaid-_r_261_ .node circle,#chatgpt-mermaid-_r_261_ .node ellipse,#chatgpt-mermaid-_r_261_ .node polygon,#chatgpt-mermaid-_r_261_ .node path{fill:rgb(0, 40,

77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_261_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_261_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_261_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_261_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_261_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter capacity and initial waterlevelEnter number of days andthresholdEnter daily inflow and outflowCalculate available waterAvailable water exceedscapacity?Calculate overflowOverflow = 0Limit available water tocapacityCheck outflowOutflow exceeds availablewater?Calculate shortage and setlevel to zeroSubtract outflow fromavailable waterCheck thresholdFirst time threshold reached?Store day numberCheck remaining daysDisplay final reportStopYesNoYesNoYesNo

8. Concepts Used from Syllabus

Syllabus concept	Application
Variables	Capacity, water level, inflow
Data types	Integer, float
Operators	Addition, subtraction, comparison
Conditional statements	Overflow and shortage checking
Loops	Multi-day simulation
Lists	Storing daily records
Tuples	Storing daily transaction details
Tuple unpacking	Extracting daily values
Algorithms	Updating water levels

9. Conclusion

The Multi-Day Water Tank Simulation program monitors water levels over multiple days. It calculates inflow and outflow, detects overflow and shortages, and identifies the first day when the water level reaches the specified threshold. This program demonstrates the practical use of Python concepts in water resource management.

Note: This is an academic simulation. Overflow is calculated before outflow, and any shortage is recorded as unmet demand rather than allowing the tank level to become negative.